

Cost of Hybrid Quantum PINNs: Parameter Count vs. Wall-Clock Time

[TODO: final title]

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Abstract

Hybrid quantum–classical physics-informed neural networks (QCPINNs) are usually reported as efficient because they use fewer trainable parameters than a classical PINN. We ask what this saving costs in wall-clock time. On a one-dimensional seawater temperature diffusion benchmark, we compare a classical PINN with four hybrid configurations that differ in qubit count and circuit depth. *Parameter count*: the quantum branch adds only 16–64 trainable parameters, and the hybrids have 2421–2717 parameters in total against 4353 for the classical baseline (a 38–44% reduction), almost all of which comes from the smaller classical part. *Timing*: [TODO: one sentence with the forward / first-order / second-order slowdown range, from final numbers] $\times$ — $\times$, growing with qubit count, circuit depth and batch size. *Takeaway*: parameter count alone overstates the efficiency of hybrid models. Accuracy is not the subject of this paper.

1 Introduction

Physics-informed neural networks (PINNs) [18] solve forward and inverse partial differential equation (PDE) problems by embedding the governing equation, boundary conditions and, for inverse problems, observations into the training loss of a neural network. We study a one-dimensional temperature diffusion problem, the seawater temperature equation of Han et al. [8], under Dirichlet, Neumann and Robin boundary conditions. [TODO: One or two sentences of application motivation, e.g. fast differentiable surrogates for energy-aware control of engineered water bodies (from the previous draft).]

Hybrid quantum–classical PINNs (QCPINNs), in which part of the network is replaced by a variational quantum circuit (VQC), have been reported to match or exceed classical PINNs while using far fewer trainable parameters [7]. Parameter count has become the standard efficiency metric in this literature.

Gap. Wall-clock cost is rarely the headline metric and, where it is reported, it is a single aggregate number (see Related work below). Yet near-term VQCs are executed by classical simulation, whose cost grows with circuit width and depth (Section 2). Training a PINN also needs first- and second-order derivatives of the network with respect to its *inputs*, which are more expensive than a plain forward pass and interact with the simulated circuit.

Research questions.

RQ1 How large is the parameter-count advantage of hybrid models, and where does it come from?

RQ2 What timing cost accompanies it, and how does that cost scale with qubit count, circuit depth, batch size and simulator backend?

Contributions.

1. A parameter-count breakdown of a classical PINN and four hybrid configurations under a common accounting.
2. Stage-wise timing of the forward pass, the first-order input gradient and the second-order input gradient.
3. Sweeps over qubit count, circuit depth, batch size and simulator backend.

Accuracy comparisons, gradient-norm ablations and barren-plateau discussion are not the subject of this paper.

Outline. [TODO: One sentence per section once the structure is final.]

Related work. Hybrid quantum PINNs have been proposed for fluid flow, wave, Helmholtz and Klein–Gordon problems [7, 12], reservoir seepage [19], hydrological forecasting [9] and Maxwell’s equations [5], with theoretical work on universal approximation and barren-plateau mitigation [11]. The headline efficiency claim in this literature is parameter count: QCPINNs are reported to reach comparable accuracy with 10–30% of the parameters of a classical PINN [7], hybrid models with fewer parameters than the best classical model [12, 22], and about 44% fewer parameters together with roughly three times fewer training epochs [9].

Cost is acknowledged but usually reported as one aggregate figure. Farea et al. [7] report a 15- to 40-fold increase in time per training iteration relative to classical PINNs, at similar memory. Leong et al. [12] state that simulating the quantum layers imposes an overhead of at least two orders of magnitude for the same number of trainable parameters, but compare training efficiency through epochs rather than wall-clock time. Trahan et al. [22] find a quantum PINN to take nearly four times as long as a classical PINN with a similar number of parameters on a 2D Poisson problem. These figures differ widely and depend on hardware, simulator, circuit and batch size; none of them, to our knowledge, breaks the cost down into forward, first-order and second-order input-gradient stages, or varies qubit count, depth, batch size and backend in a controlled way. On the simulation side, Chen et al. [5] develop a GPU-accelerated simulation library for computing circuit outputs and their derivatives, so as to make QPINN training practical. This paper fills the gap with a controlled timing study next to the parameter count. [TODO: Read the full texts of [5, 9, 11, 19]; only their abstracts were checked. Confirm the “to our knowledge” claim with a broader literature search, and check the journal version of [12] (Computers & Fluids) for its own timing numbers.]

2 Background and Key Concepts

2.1 PINN loss structure

We consider the diffusion equation

$$\frac{\partial T}{\partial t} = k_z \frac{\partial^2 T}{\partial z^2}, \quad z \in [0, 2], \quad t \in [0, 5], \quad (1)$$

with diffusivity $k_z = 0.05$ in forward problems and k_z estimated from data in inverse problems. Let $\hat{T}_\theta(z, t)$ be the network output with trainable parameters θ and input $x = (z, t)$. The PINN loss combines four mean-squared terms,

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{PDE}} + \mathcal{L}_{\text{BC}} + \mathcal{L}_{\text{IC}} + \mathcal{L}_{\text{data}}, \quad (2)$$

namely the PDE residual at interior collocation points, the boundary-condition residual, the misfit to the initial condition, and, for inverse problems only, the misfit to observations; all terms have unit weight.

Initial and boundary conditions. Closed-form solutions exist for all three boundary-condition types. Table 1 lists the conditions imposed. The initial condition is the exact solution at $t = 0$; for the Dirichlet problem the exact solution vanishes at both boundaries, so $\mathcal{L}_{\text{BC}}$ penalises $\hat{T}_\theta(0, t)$ and $\hat{T}_\theta(2, t)$ directly. The Neumann and Robin conditions involve $\partial\hat{T}_\theta/\partial z$ at the boundaries and therefore already require a first-order input gradient (Section 2.2).

Table 1: Initial and boundary conditions of the three benchmark problems. Each boundary condition holds at $z = 0$ and $z = 2$ for all $t \in [0, 5]$.

Type	Boundary condition	Initial condition $T(z, 0)$
Dirichlet	$T = 0$	$\sin \frac{\pi z}{2} + \sin \pi z$
Neumann	$\partial T/\partial z = 0$	$\cos \frac{\pi z}{2} + \cos \pi z$
Robin	$\partial T/\partial z + T = 0$	$e^{-z} + \frac{\pi}{2} \cos \frac{\pi z}{2} - \sin \frac{\pi z}{2} + \pi \cos \pi z - \sin \pi z$

Training points. We sample 2540 interior collocation points uniformly in $[0, 2] \times [0, 5]$, 40 points on each of the two boundaries, and 160 points at $t = 0$. Inverse problems additionally use 12 observations on a uniform 4×3 grid in (z, t) .

2.2 First-order input gradient

The PDE residual is

$$r_\theta(z, t) = \frac{\partial \hat{T}_\theta}{\partial t} - k_z \frac{\partial^2 \hat{T}_\theta}{\partial z^2}. \quad (3)$$

Evaluating it requires the *first-order gradient* of the output with respect to its input, $\nabla_x \hat{T}_\theta = (\partial \hat{T}_\theta/\partial z, \partial \hat{T}_\theta/\partial t)$, which PINNs obtain by automatic differentiation of the network [1, 13, 18]. It supplies the time derivative in r_θ and the boundary residuals of the Neumann ($\partial \hat{T}_\theta/\partial z = 0$) and Robin ($\partial \hat{T}_\theta/\partial z + \hat{T}_\theta = 0$) problems. It must be distinguished from the *parameter gradient* $\partial \mathcal{L}/\partial \theta$ used by the optimiser, which is a further backward pass through the graph of the input gradients.

2.3 Second-order input gradient

The diffusion term requires $\partial^2 \hat{T}_\theta/\partial z^2$, obtained by differentiating $\partial \hat{T}_\theta/\partial z$ once more with respect to z . We compute both derivatives by reverse-mode automatic differentiation [1] in PyTorch [16], retaining the computation graph after the first pass so that it can be differentiated again; differentiating through a gradient with respect to the input in this way is known as double backpropagation [6]. The second-order gradient therefore costs a second backward pass through the graph created by the first, on top of the forward pass and the first-order pass.

2.4 Variational quantum circuits

A VQC acts on n qubits and consists of an encoding of classical features into gate angles followed by L layers of parametrised single-qubit rotations and entangling gates; its outputs are expectation values of observables [2, 4]. The trainable quantum parameters are the rotation angles; n is the *width* and L the *depth* of the circuit. The encoding gates determine which functions of the input the circuit can represent [21]; we use angle embedding (Section 3). VQCs are the model class of the noisy intermediate-scale quantum (NISQ) era [17], but in practice they are mostly run on classical simulators, and so are ours; we do not use quantum hardware. A statevector simulator stores 2^n complex amplitudes and, in the worst case, applies each gate in $O(2^n)$ operations, so memory and time grow exponentially with n and linearly with the number of gates, hence with the depth L [23].

2.5 How circuit gradients are obtained

On a classical simulator, as in our implementation, gradients are obtained by backpropagation: each backward pass traverses the operations that simulate the circuit, so the cost of every stage grows with the simulated circuit. On quantum hardware this is not possible, because intermediate quantum states cannot be measured and stored for reuse without destroying them [20]. Gradients of circuit outputs are instead obtained with the parameter-shift rule [15, 20]: for a gate generated by a Pauli operator, the derivative with respect to its angle is computed from two evaluations of the circuit with that angle shifted. Higher-order derivatives follow by iterating the rule, with up to 2^d shifted evaluations per derivative entry of order d (four for the Hessian) [14]. The same rule applies to the encoding gates, which gives derivatives of a quantum model with respect to its input [10]. In our hybrid network the circuit input is produced by classical layers, so input derivatives of the full network combine the chain rule through those layers with derivatives of the circuit. We measure the simulator case only and make no claim that simulator ratios carry over to hardware.

2.6 Notation

Table 2: Notation and terminology.

Symbol / term	Meaning
$x = (z, t)$	network input: depth and time
$\hat{T}_\theta$	network output, temperature, with parameters θ
k_z	thermal diffusivity
$\nabla_x \hat{T}_\theta$	first-order input gradient
$\partial^2 \hat{T}_\theta / \partial z^2$	second-order input gradient
$\partial \mathcal{L} / \partial \theta$	parameter gradient (optimiser; not a timed stage)
n, L	number of qubits (width), number of circuit layers (depth)
forward / first_grad / second_grad	the three cumulative timed stages (Section 4)

3 Model Architectures

3.1 Classical baseline

The classical PINN is a fully connected network with five hidden layers of width 32 and tanh activations, mapping $(z, t) \mapsto \hat{T}_\theta$; it has 4353 trainable parameters.

n = number of qubits, L = number of variational layers; tensors between blocks have shape $[\text{batch}, n]$

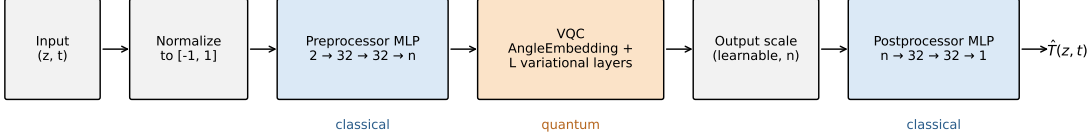


Figure 1: Hybrid QCPINN: classical layers, quantum layer, classical layers.

Table 3: Configurations. Quantum parameters are $4nL$; the classical share includes the n learnable output scales.

Config	Qubits n	Layers L	Quantum	Classical	Total	Ratio to classical
classical	—	—	0	4353	4353	1.00
narrow	2	2	16	2405	2421	0.56
baseline	4	2	32	2537	2569	0.59
wide	6	2	48	2669	2717	0.62
deep	4	4	64	2537	2601	0.60

3.2 Hybrid architecture

The hybrid network (Figure 1) replaces the middle of the network with a VQC: a classical preprocessor $2 \rightarrow 32 \rightarrow 32 \rightarrow n$ (tanh activations) maps the normalised input to n features, the VQC returns n Pauli- Z expectation values, a learnable per-qubit scale is applied, and a classical post-processor $n \rightarrow 32 \rightarrow 32 \rightarrow 1$ produces $\hat{T}_\theta$.

Circuit. The features are encoded by angle embedding (one R_X per qubit). Each of the L layers applies R_Z, R_X on every qubit, a ring of CNOT gates $i \rightarrow (i + 1) \bmod n$, then R_X, R_Z on every qubit, giving $4nL$ trainable quantum parameters. The output is $\langle Z_i \rangle$ for each qubit. The circuit follows the design of Farea et al. [7]. Gradients are obtained by backpropagation on the `default.qubit` simulator (PennyLane [3]).

3.3 Configurations

Table 3 lists the four hybrid configurations. *Wide* increases the qubit count of the baseline from 4 to 6 at fixed depth; *narrow* reduces it to 2; *deep* keeps 4 qubits and doubles the depth to 4 layers. Figure 2 shows an exemplar circuit for each. [TODO: Confirm that “deep” = 4 qubits, 4 layers is what the paper should call deep.]

4 Experimental Setup

4.1 Timed stages

We time three cumulative stages on the same batch of collocation points:

1. **forward**: evaluate $\hat{T}_\theta(x)$;
2. **first_grad**: the forward pass plus $\nabla_x \hat{T}_\theta$;

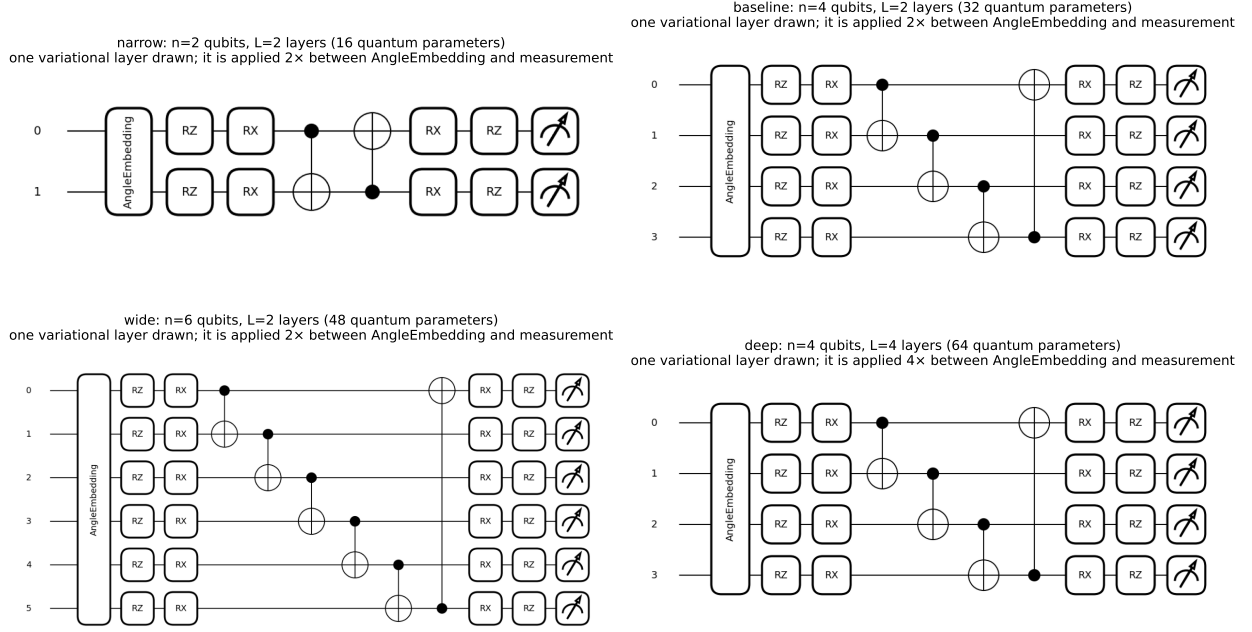


Figure 2: Exemplar circuits of the four hybrid configurations (narrow, baseline, wide, deep).

3. **second_grad:** the forward pass, $\nabla_x \hat{T}_\theta$ and $\partial^2 \hat{T}_\theta / \partial z^2$.

The stages are cumulative, so the *second-over-forward ratio* is the cost of everything needed for the PDE residual relative to a plain evaluation. The parameter gradient $\partial \mathcal{L} / \partial \theta$ is not part of these stages, so a full training step costs more than the second_grad time. [TODO: Decide whether to keep this three-stage definition. The latest timing script also times a fourth stage, **param_grad**, i.e. the optimiser’s own backward pass; if it goes into the paper, update this subsection, Table 2 and Section 2.2.]

4.2 Protocol

Each stage is timed over [TODO: 20] repetitions after [TODO: 3] warm-up calls, on a batch of [TODO: 2540] collocation points sampled uniformly in $[0, 2] \times [0, 5]$, matching the training batch size. Network weights, batch data and circuit parameters are redrawn for each of 3 independent seeds; we report the mean and the standard deviation across seeds. Backends are **default.qubit** (vectorised over the batch) and **lightning.qubit**. [TODO: Confirm all numbers above against **qcpinn_timing_profile.py** (**N_REPS**, **N_WARMUP**, **BATCH**, **SWEEP_SEEDS**) before submission.]

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